

Computer Networks and the Internet: An Overview

Chapter 1

Roadmap

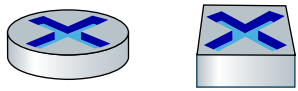
- Internet structure
 - *overview*
 - *protocols*
 - *core, edge, access*
- Packet vs circuit switching
- Performance metrics
 - *loss, delay, throughput*
- Internet layers

The Internet: "a network engineer" view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers, switches*

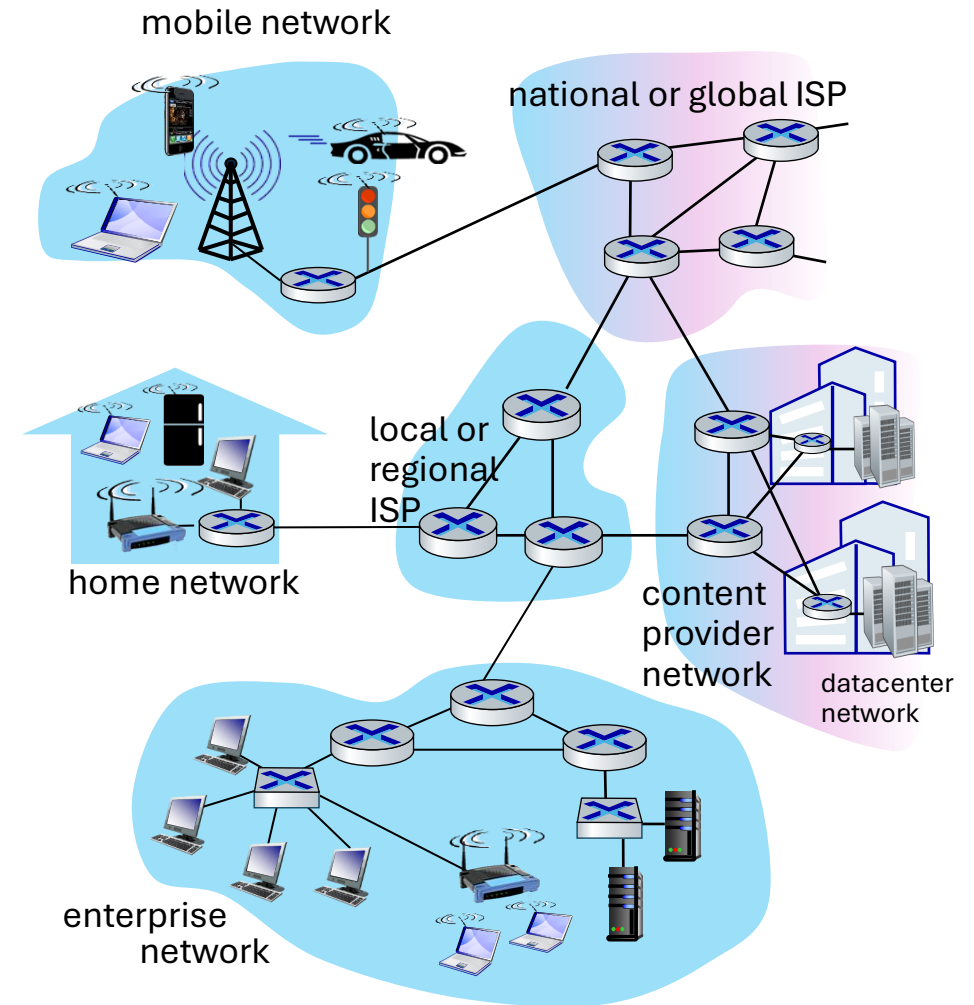
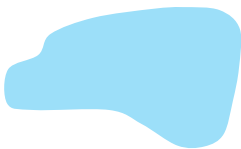


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

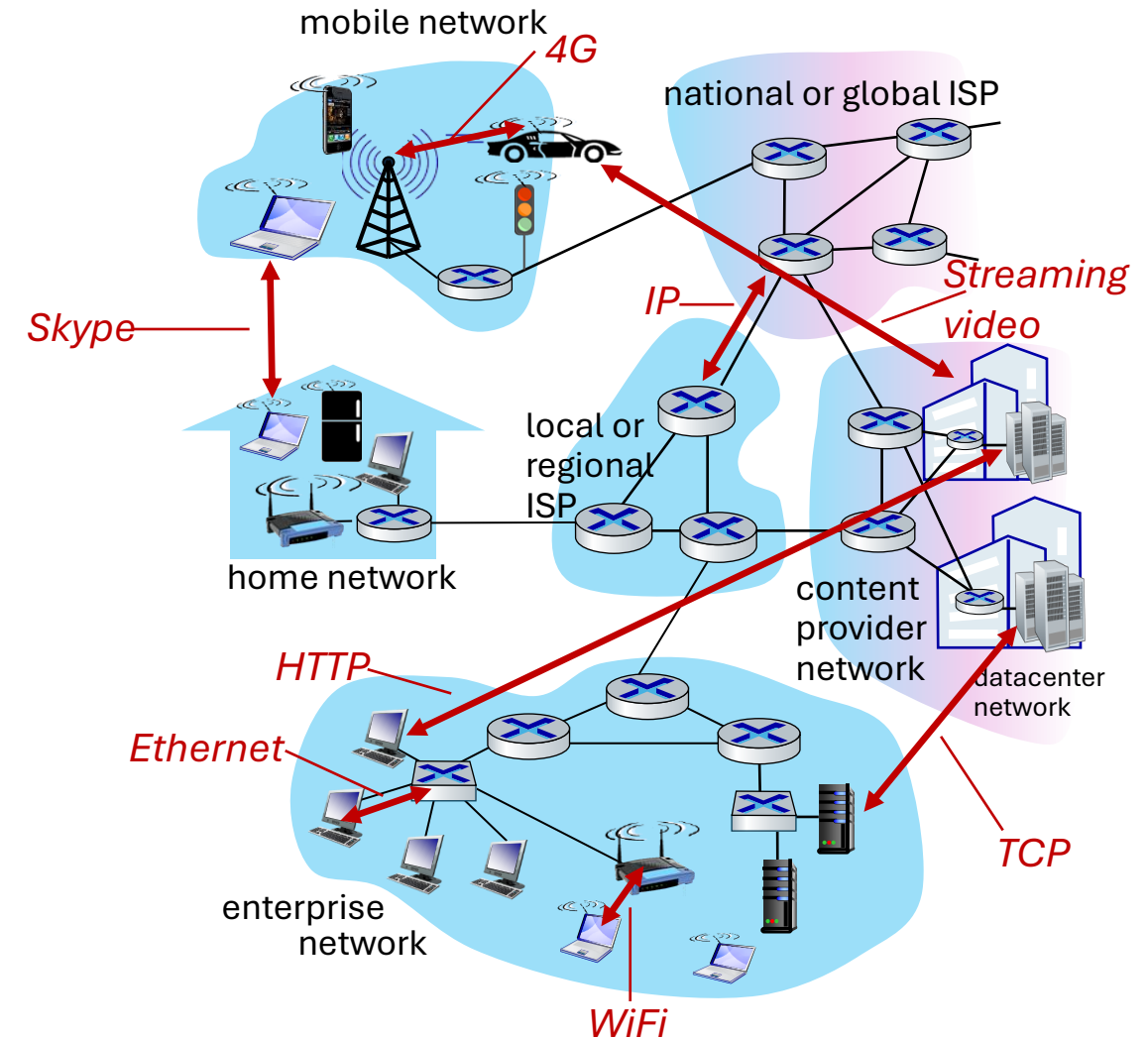
Networks

- collection of devices, routers, links: managed by an organization



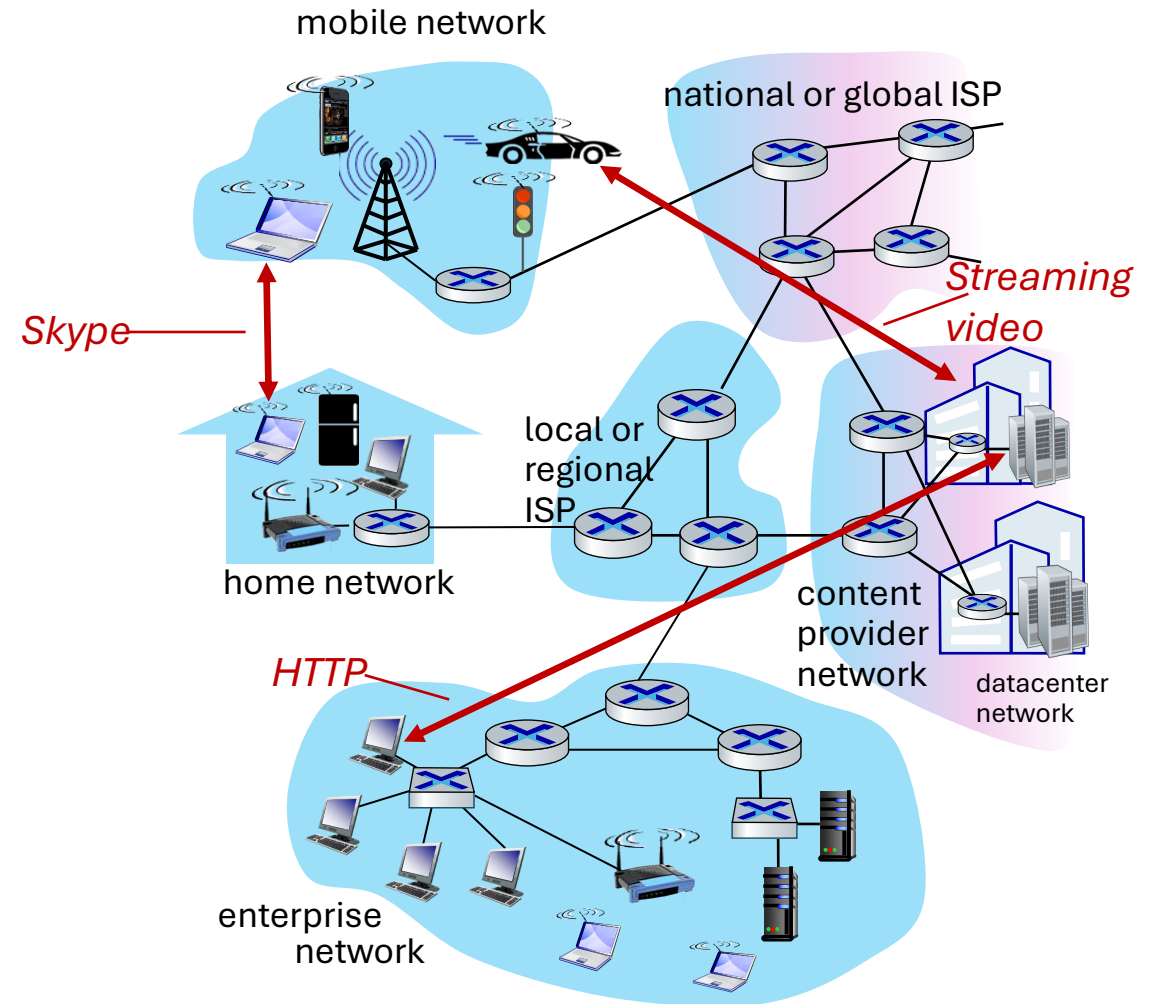
The Internet: "a network engineer" view

- "network of networks"
- **protocols** are everywhere:
 - control sending, receiving of messages
 - e.g., HTTP, Skype, TCP, IP, WiFi, 4G/5G, Ethernet
- **standards**
 - protocol ~ algorithm
 - standard ~ template, objectives
 - important standardization bodies: RFC, IETF, IEEE



The Internet: "a software engineer" view

- **Infrastructure** that provides services to applications:
 - web, streaming, email, games, social media...
- Provides **programming interface** to distribute applications:
 - "hooks" allowing apps to send/receive messages (e.g., socket programming)
 - "service options," analogous to postal service (e.g., reliable vs unreliable data transfer)



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

... specific messages sent
... specific actions taken
when message received,
or other events

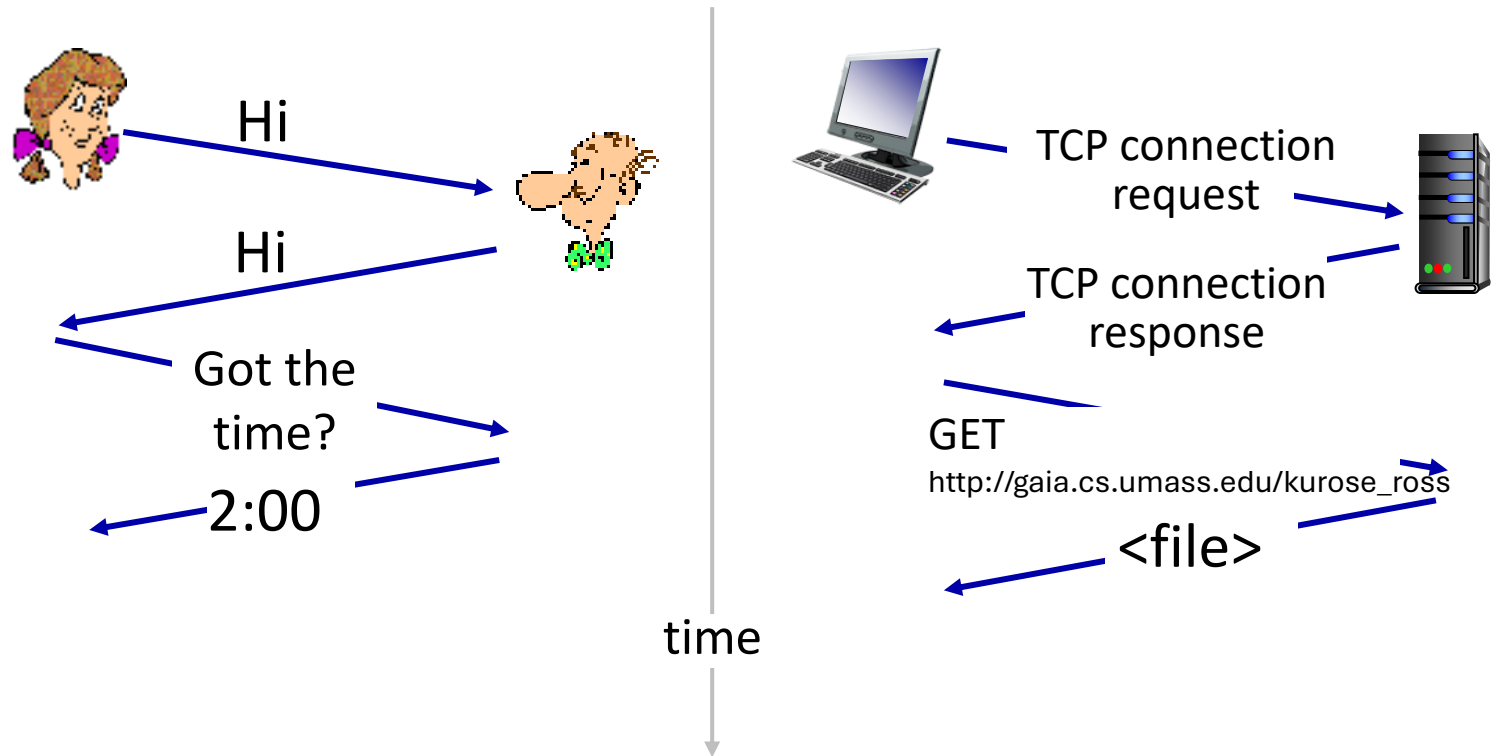
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

*Protocols define the **format, order** of **messages sent and received** among network entities, and **actions taken** on message transmission, receipt*

What's a protocol?

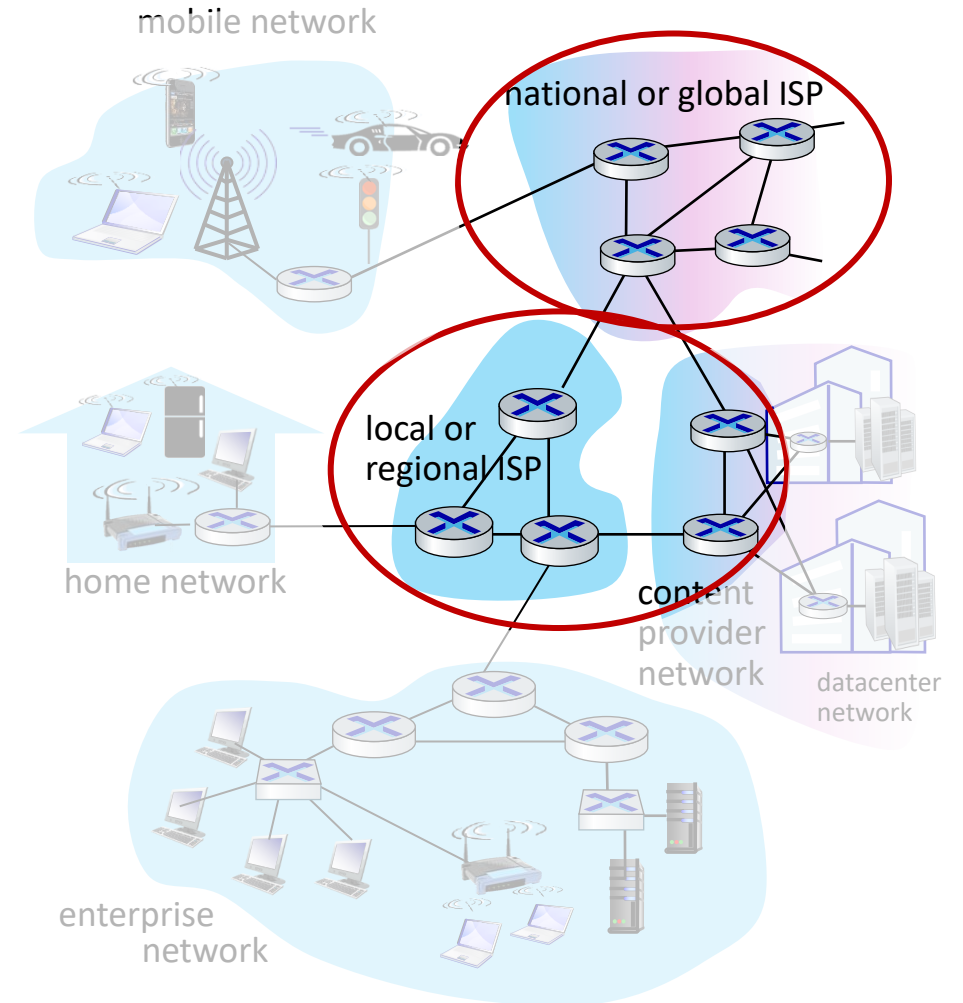
A human protocol and a computer network protocol:



Internet structure

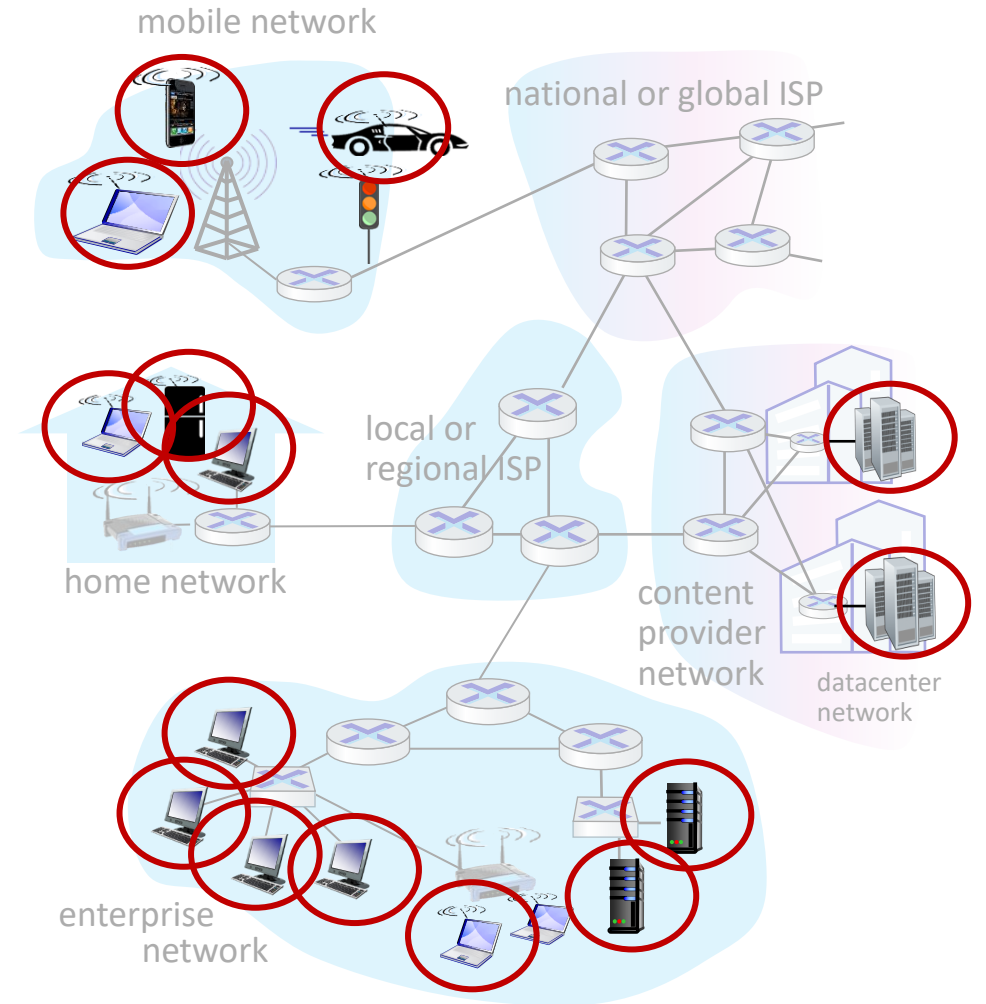
- **network core:**
 - mesh of interconnected *routers*, *networks*
- **network edge:**
 - hosts: clients and servers
 - servers often in data centers
- **access network:**
 - wired, wireless links
 - often used to describe *last-mile* links (e.g., WiFi, 4G/5G, DSL)

Definition: **ISP** (Internet Service Provider): entity/organization that manages a given network



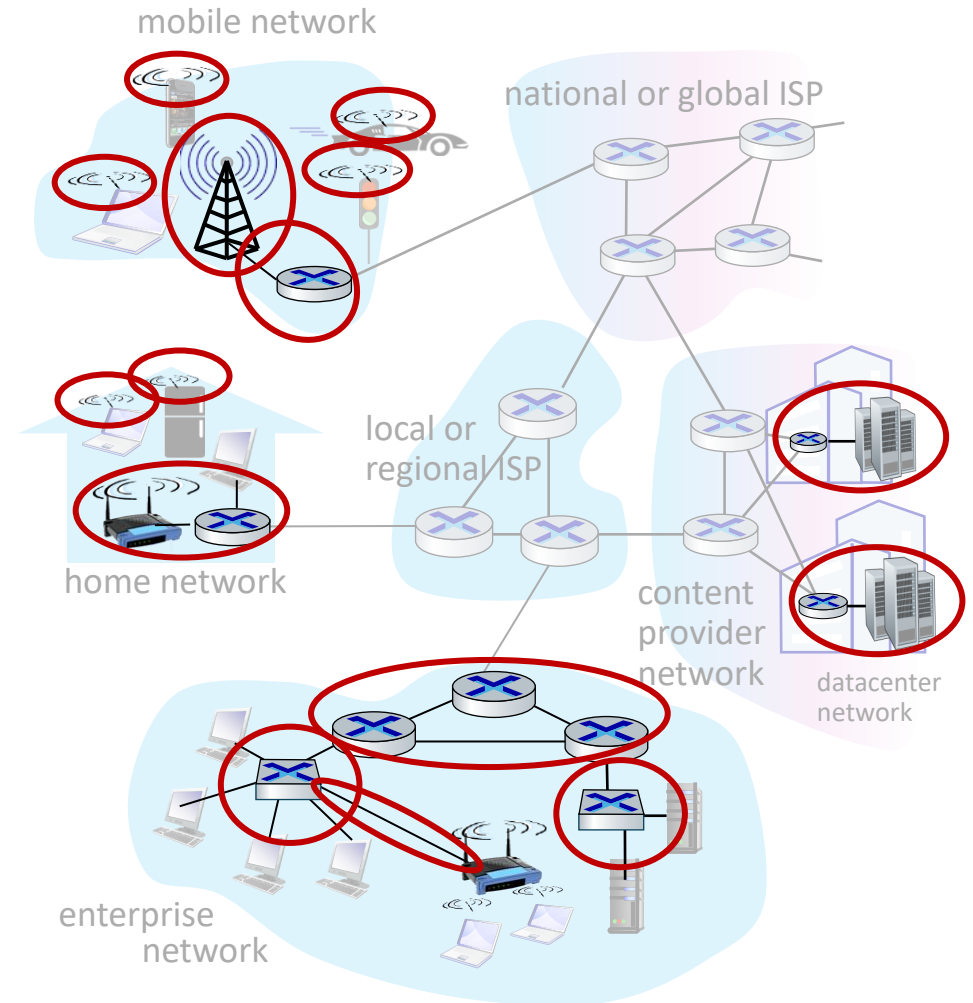
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Internet structure

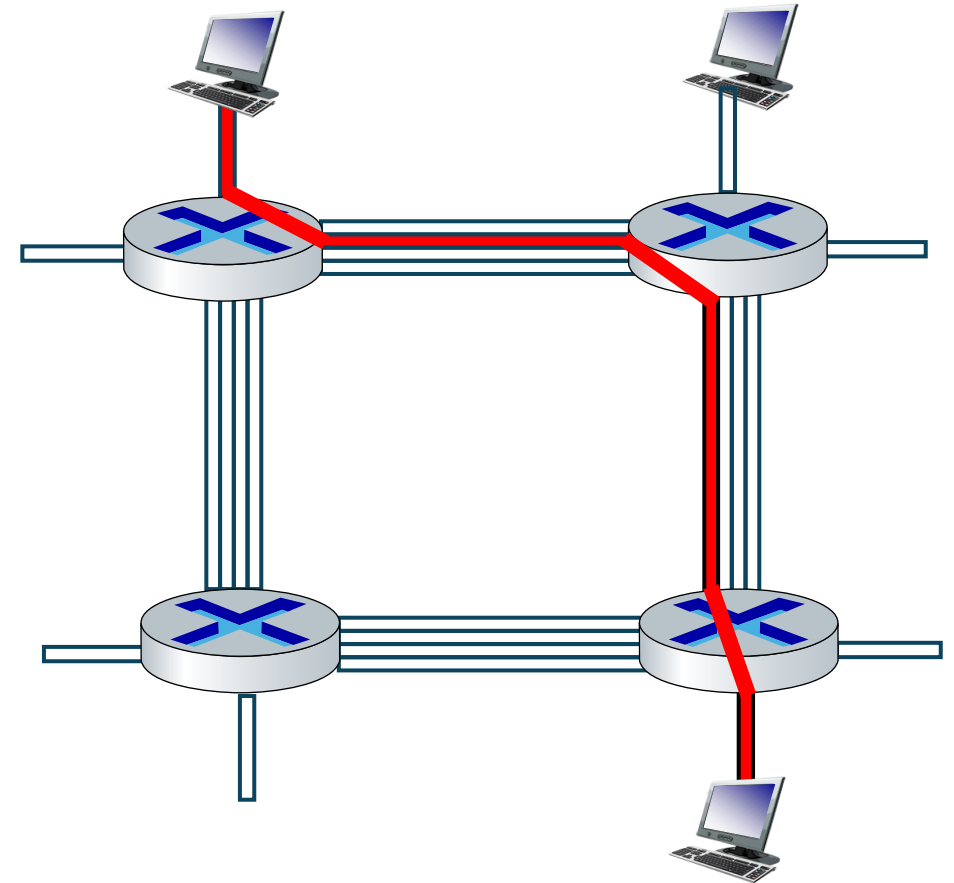
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Data transfer: Two broad approaches

1- Circuit switching:

- resources **reserved** for "call" between source and destination
- pre-defined route to destination
- guaranteed performance
- data often transmitted in **streams**



How to *reserve* resources?

Old telephone systems:

- telephone operator physically interconnected source and destination
- *reserved, exclusive* link between end devices

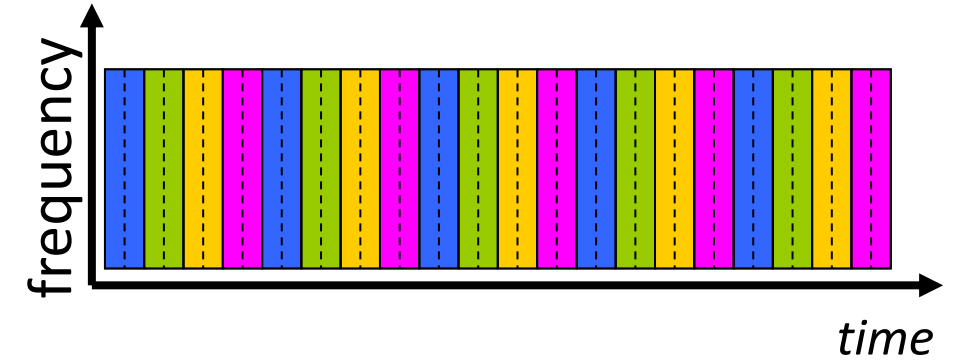
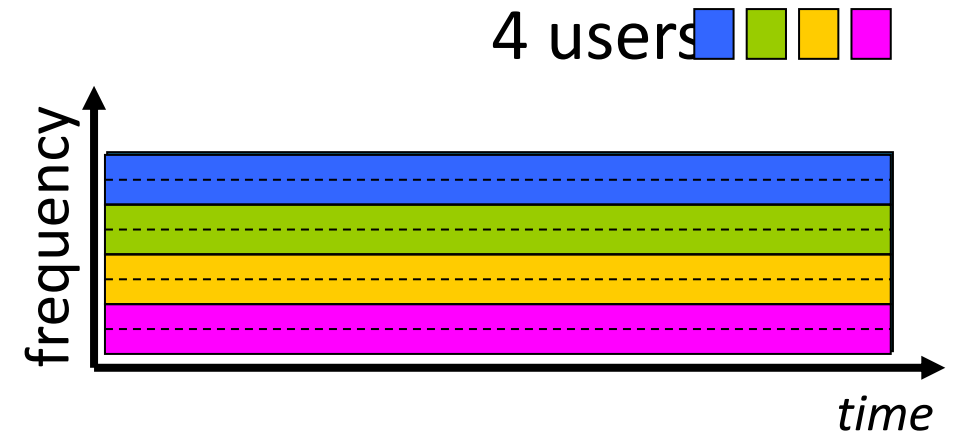


Telephone exchange

How to *reserve* resources?

Modern approaches:

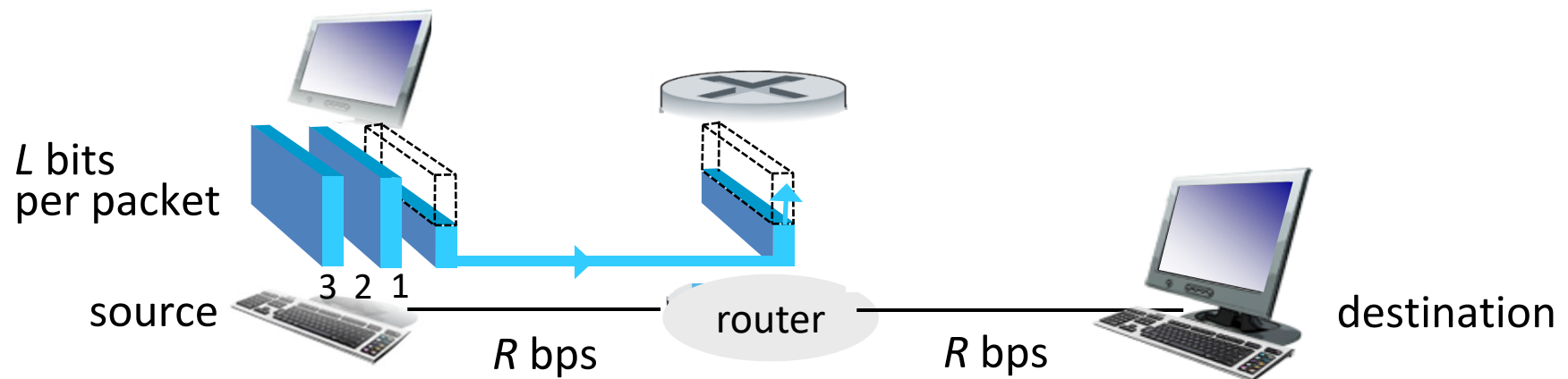
- Frequency Division Multiplexing (FDM)
 - optical, electromagnetic frequencies divided into narrow frequency bands
 - each call/user allocated its own band
- Time Division Multiplexing (TDM)
 - time divided into slots
 - each call/user allocated periodic slots



Data transfer: Two broad approaches

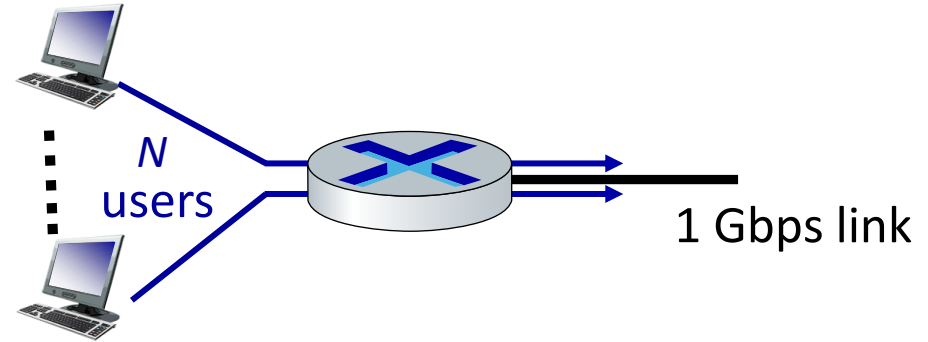
2- Packet switching: store-and-forward

- hosts break messages into **packets**
- packets are *stored and forwarded* one at a time (e.g., first-in, first-out)
- no resource reservation; no pre-defined route to destination
- no performance guarantees



Circuit switching vs Packet switching

- 1 Gbps link
- each user:
 - needs 100 Mbps when "active"
 - active 10% of time

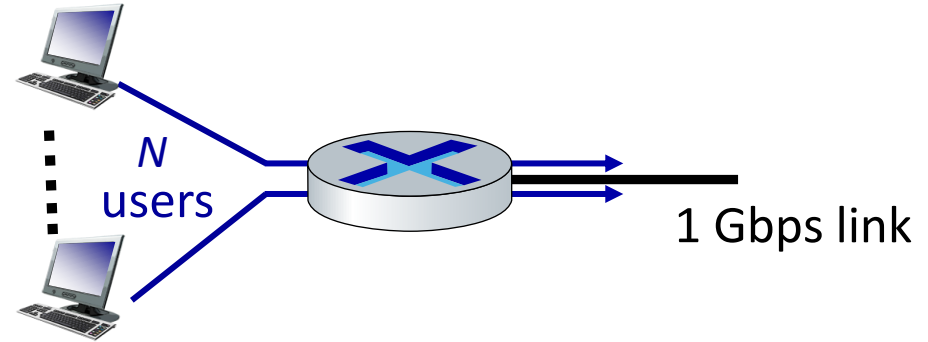


Q: How many users can be served using **circuit switching**?

A: 10 users

Circuit switching vs Packet switching

- 1 Gbps link
- each user:
 - needs 100 Mbps when "active"
 - active 10% of time



Q: How many users can be served using **packet switching**?

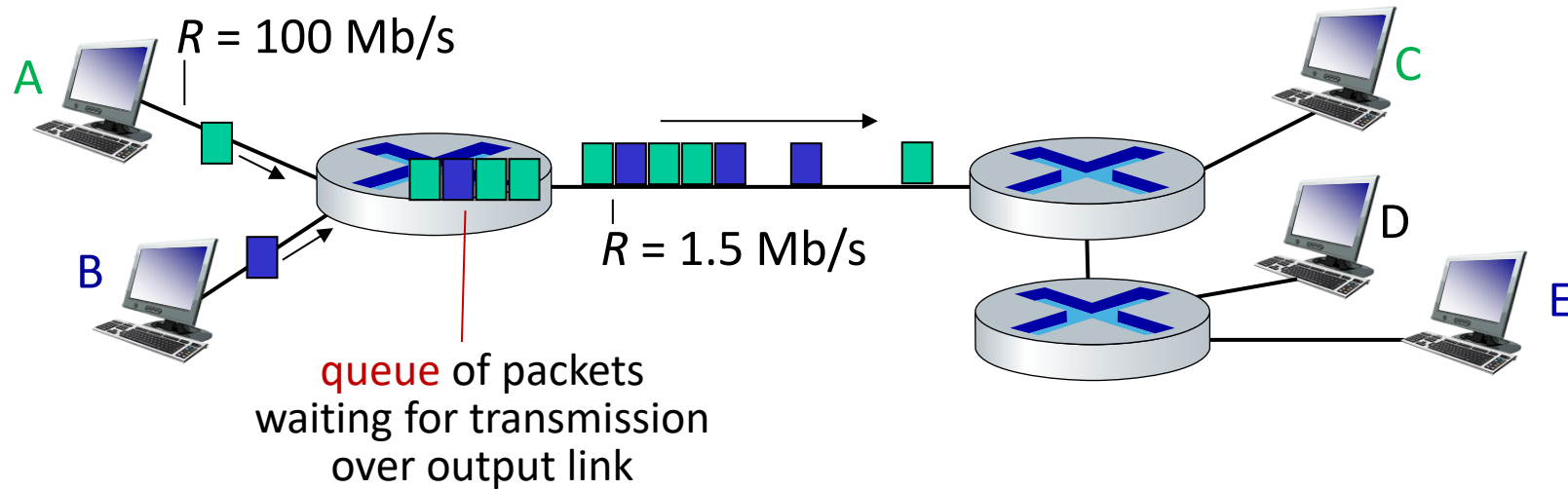
A: with 35 users, probability > 10 active users at same time is less than 0.004.

Is packet switching a winner?

The Internet uses packet switching.

Pros	Cons
• great for bursty data (sometimes has data to send, sometimes does not) • resource sharing • simple (no "call" setup)	• excessive congestions possible • no performance guarantees

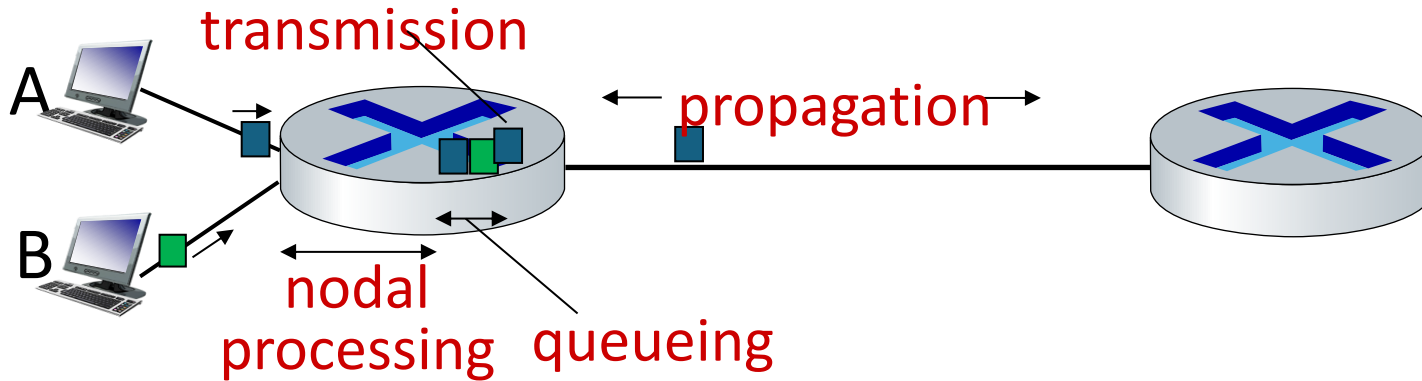
Packet switching: Queueing



What if more than one user is active at a time? If arrival rate (in bps) to link exceeds transmission rate (bps) of link for some time:

- packets will **queue**, waiting to be transmitted on output link
- packets can be **dropped** (lost) if memory (buffer) in router fills up

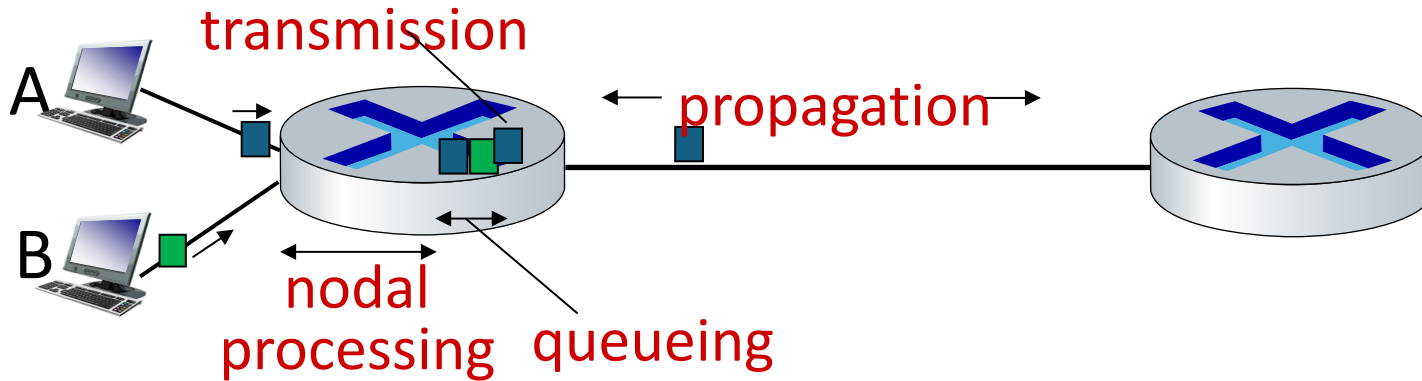
How packet delay occurs?



$$\text{delay} = d_{trans} + d_{prop} + d_{proc} + d_{queue}$$

- **transmission delay:** rate information is transmitted to link
 - L: packet size (bits)
 - R: link transmission rate (bps)
 - $d_{trans} = L/R$

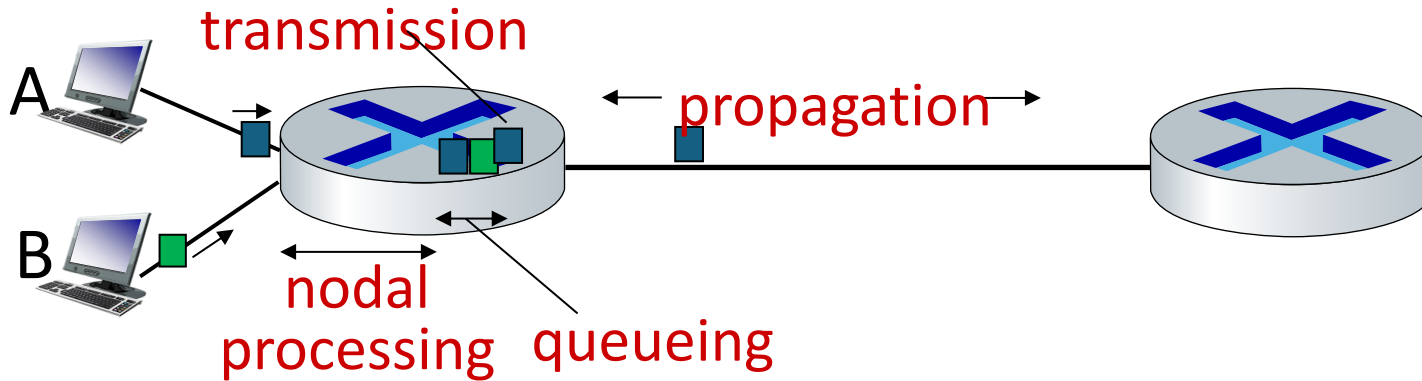
How packet delay occurs?



$$\text{delay} = d_{trans} + d_{prop} + d_{proc} + d_{queue}$$

- **propagation delay:** travel time
 - d : length of physical link
 - s : propagation speed ($\sim 2 \times 10^8$ m/s)
 - $d_{prop} = d/s$

How packet delay occurs?



$$\text{delay} = d_{trans} + d_{prop} + d_{proc} + d_{queue}$$

- **processing delay:**

- time to process packet headers
- time to check if packet is not corrupted
- function of "CPU power"

Queueing delay

- depends on in and out rates
- harder to estimate exactly

Useful metric: **traffic intensity**

- α : average packet arrival rate (pkt/s)
- L : packet length (bits)
- R : link transmission rate (bps)

$$\frac{L \times \alpha}{R} : \frac{\text{arrival rate}}{\text{service rate}}$$



$La/R \ll 1$

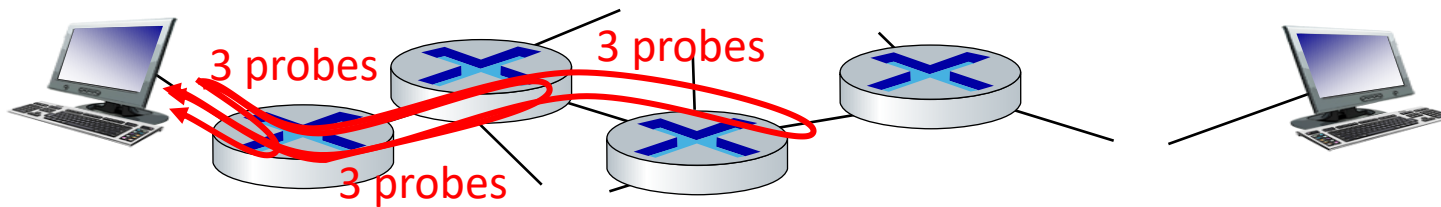
$La/R \rightarrow 1$

- $La/R \ll 1$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay high (**Q**: what happens to "excess" packets?)

"Real" Internet delays and routes

traceroute program:

- sends 3 packets that will reach router i on path towards destination
- router i will return packets to sender
- sender measures time interval between transmission and reply



Real Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

looks like delays
decrease! Why?

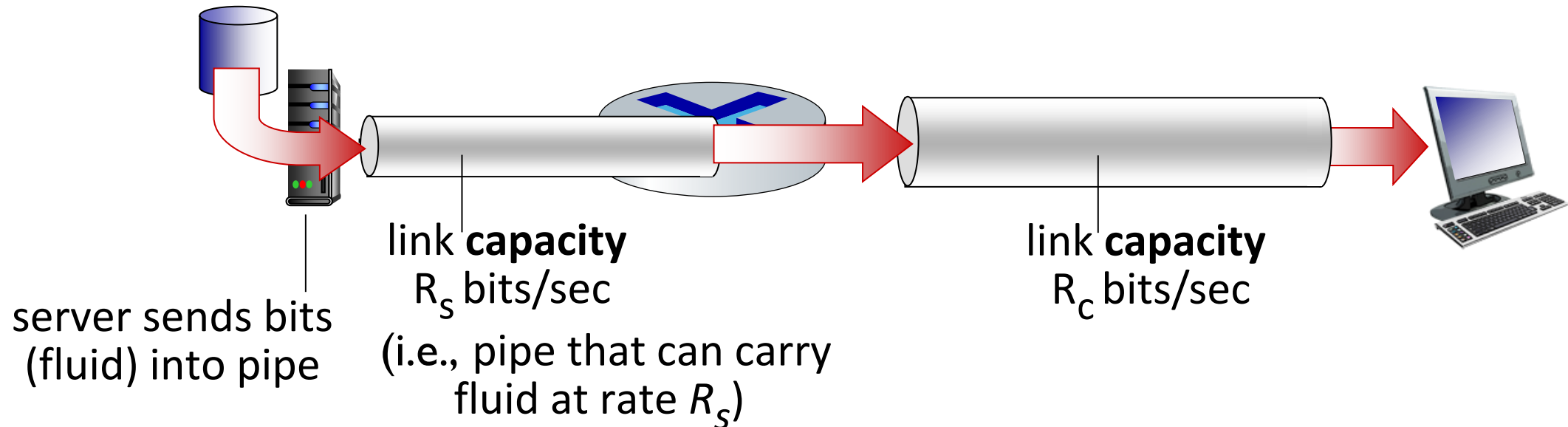
* means no response (probe lost, router not replying)

hop	hostname (ip)	1st probe	2nd probe	3rd probe
1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms
4	jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms
5	jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms
17	***			
18	***			
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms

* Do some traceroutes from exotic countries at www.traceroute.org

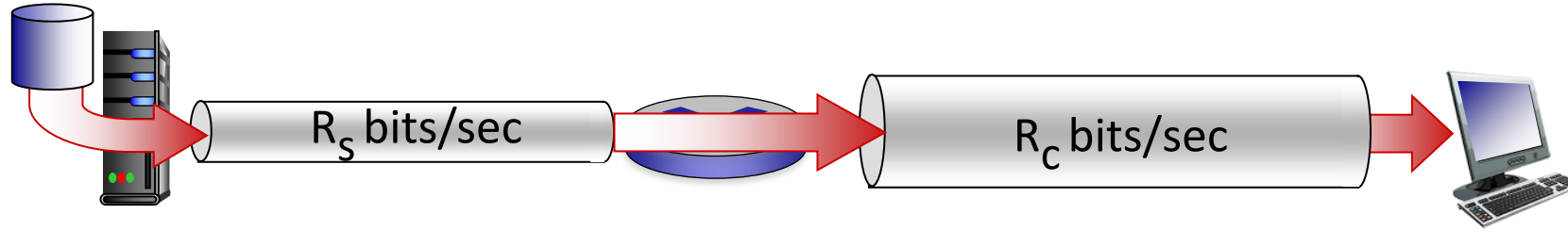
Throughput

- *average* rate (bits/time unit) at which bits are being sent from sender to receiver
- analogy: water pipes

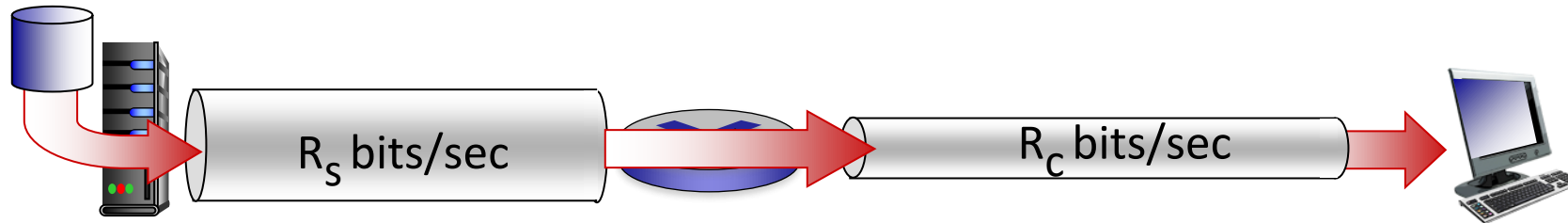


Throughput

$R_s < R_c$ What is average end-end throughput?



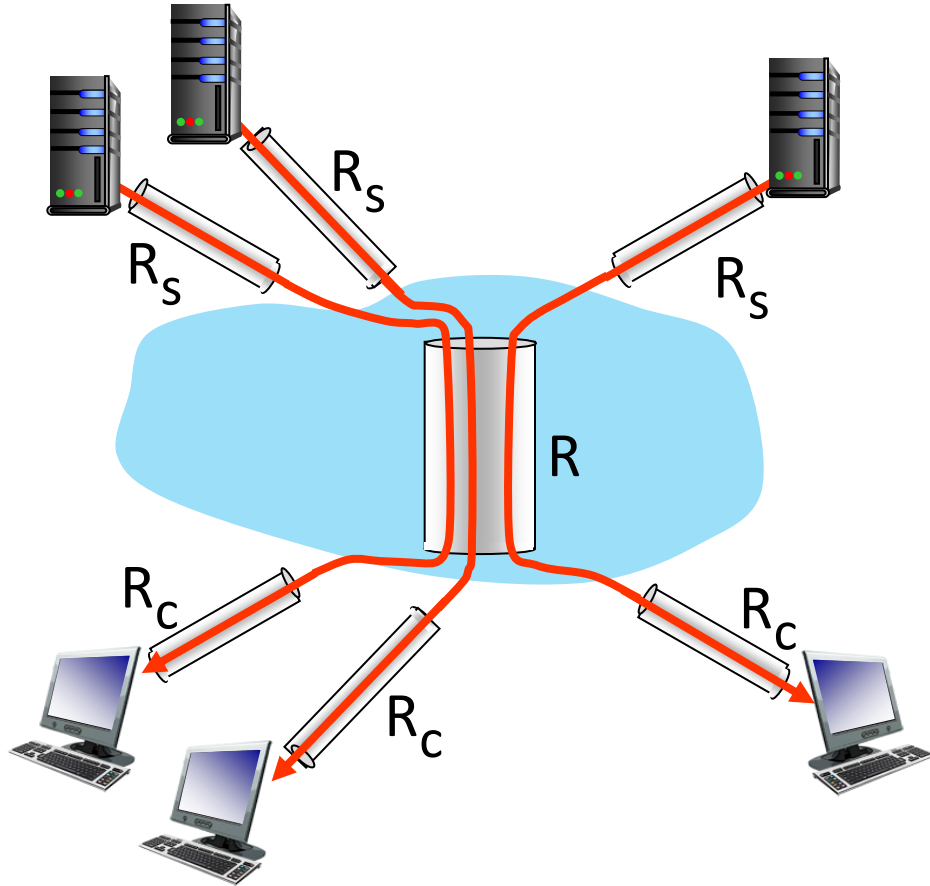
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: network scenario

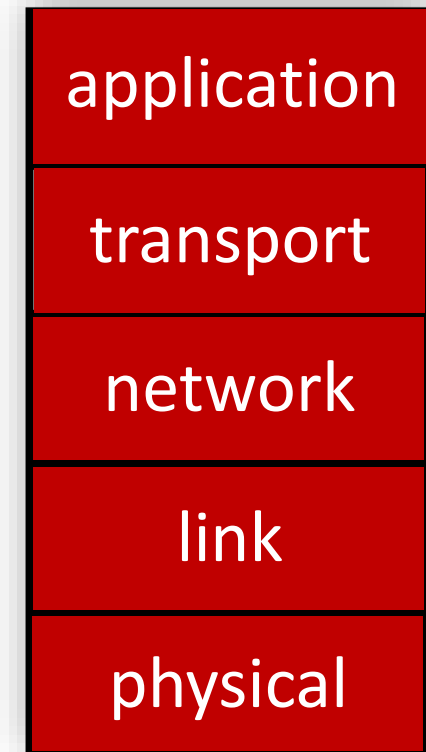


10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

Layered Internet protocol stack

- *application*: supporting network applications
 - HTTP, IMAP, SMTP, DNS
- *transport*: **process-process** data transfer
 - TCP, UDP
- *network*: routing of datagrams from **source to destination**
 - IP, routing protocols
- *link*: data transfer between **neighboring network elements**
 - Ethernet, 802.11 (WiFi), PPP
- *physical*: bits “on the wire”



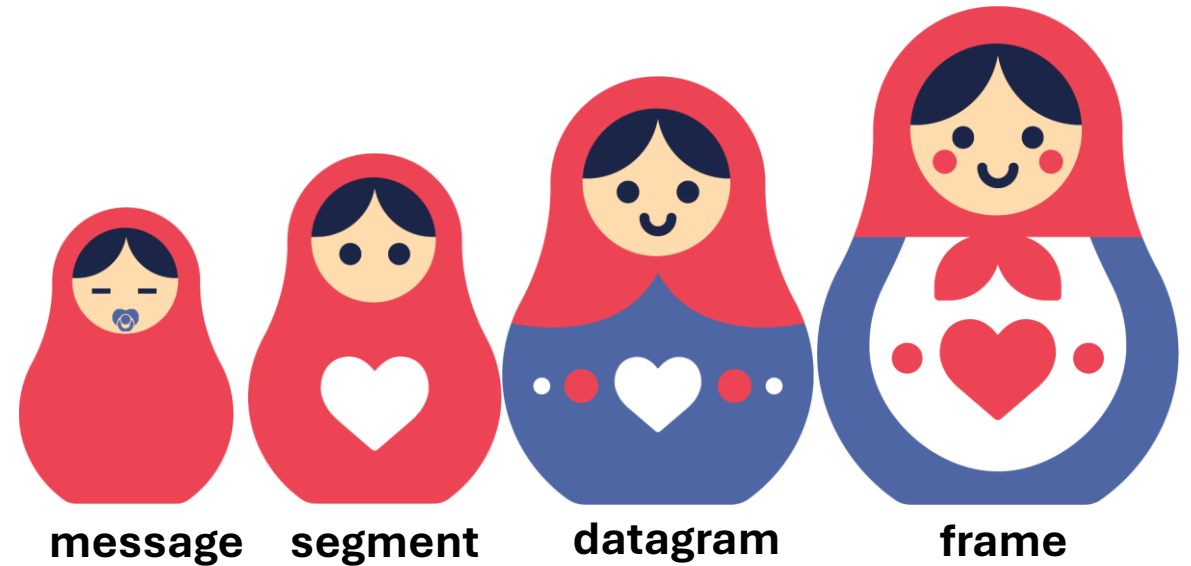
Why layering?

Layering is a common approach to designing complex systems. The explicit, modular structure

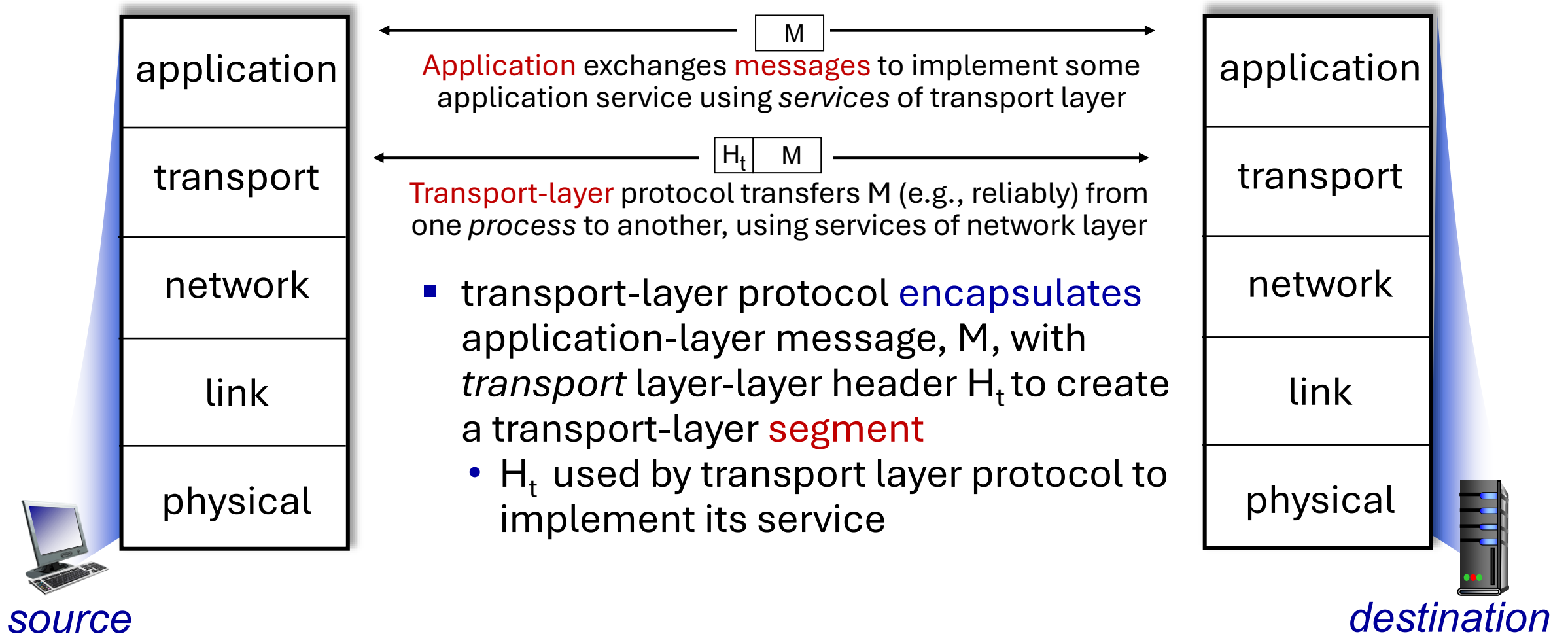
- allows identifying relationship of system's pieces
- eases maintenance, updating system

Encapsulation

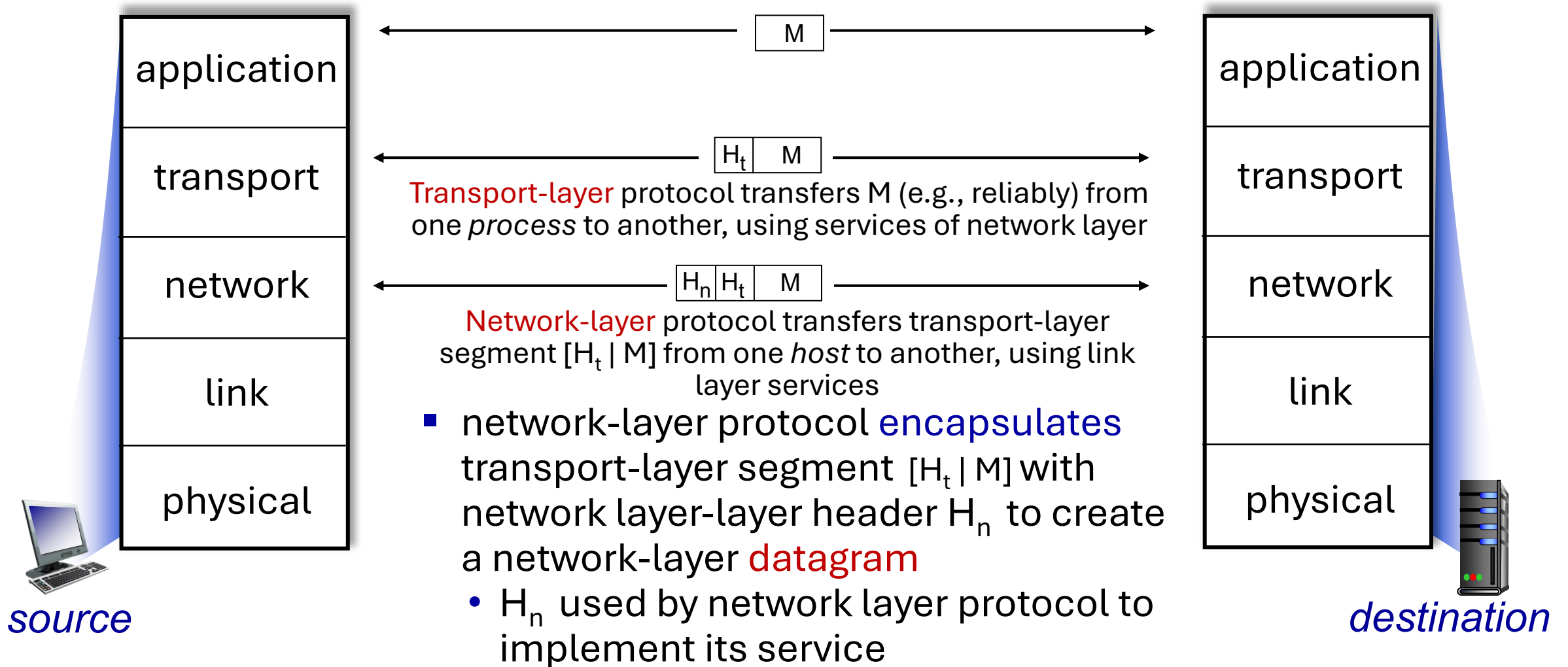
Matryoshka dolls (stacking dolls)



Services, Layering and Encapsulation



Services, Layering and Encapsulation



Services, Layering and Encapsulation

